

Risk-Budgeted Imitative Reinforcement Learning for Multi-Level Market Making

Inventory-risk control through explicit constraints

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Talk Outline

Part I introduces the market-making setup and the unconstrained IMM agent. Part II starts on slide 28 and develops explicit risk budgets, distributional learning, and research extensions.

Market-making setup

1. **Limit order book**
prices, queue sizes, and order flow
2. **Liquidity provision**
spread capture creates inventory risk
3. **Sequential decision problem**
MDP, reward, value, and Bellman relation

From PPO to IMM

4. **Previous PPO baseline**
fixed catalogue of quote configurations
5. **Continuous multi-level actions**
spreads, sizes, and reservation-price shift
6. **IMM with TD3**
predictive representation and imitation loss

Limit Order Books: Prices, Volumes, and Order Flow

At each decision time the policy sees a finite-depth snapshot of buy and sell liquidity.

$$L_t = \left(\{(p_{t,i}^b, v_{t,i}^b)\}_{i=1}^D, \{(p_{t,i}^a, v_{t,i}^a)\}_{i=1}^D \right)$$

Best quotes and spread

$$m_t = \frac{p_{t,1}^a + p_{t,1}^b}{2}, \quad s_t = p_{t,1}^a - p_{t,1}^b.$$

Book event

$$L_{t+1} = \mathcal{T}(L_t, \varepsilon_{t+1}),$$

ε_{t+1} collects submissions, executions, and cancellations.

Qualification The observation is a finite-depth, discrete-time projection of the market, not its complete state.

Market Making as Liquidity Provision

A market maker posts both sides of the book. Profit appears only if executions are balanced.

$$\tilde{p}_t^b = m_t - \delta_t^b, \quad \tilde{p}_t^a = m_t + \delta_t^a$$

Both sides fill

$$R_t^{\text{spread}} = u(\delta_t^a + \delta_t^b) > 0.$$

Spread is captured.

Only one side fills

$$z_{t+1} = z_t + e_t^b - e_t^a \neq 0.$$

Inventory remains exposed.

Key point The action affects immediate spread revenue and the risk carried into future decisions.

Avellaneda and Stoikov (2008), *High-frequency trading in a limit order book*.

Inventory Risk and Mark-to-Market Wealth

Mark-to-market wealth separates earned spread from the price risk of the remaining position.

$$W_t = x_t + z_t m_t, \quad \text{PnL}_{0:T} = W_T - W_0$$

One-step decomposition

$$\Delta W_t = \underbrace{\delta_t^b e_t^b + \delta_t^a e_t^a}_{\text{spread capture}} + \underbrace{z_{t+1} \Delta m_t}_{\text{inventory revaluation}} - \underbrace{f_t}_{\text{fees}}$$

Key point Inventory risk enters through $z_{t+1} \Delta m_t$: the same price move matters more when the position is larger.

Market Making as a Sequential Decision Process

A quote affects both current spread revenue and the inventory carried into future decisions.

$$\mathcal{M} = (\mathcal{X}, \mathcal{A}, P, r, \gamma)$$

$$X_t \xrightarrow{O} S_t \xrightarrow{\pi_\theta} a_t, \quad (r_t, X_{t+1}) \sim P(\cdot \mid X_t, a_t)$$

Policy view

$S_t = O(X_t)$ is the information available to the agent; the policy chooses a_t from this observation.

Finite-horizon objective

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k r_{t+k}, \quad \max_{\theta} J(\pi_\theta) = \mathbb{E}[G_0].$$

Key point RL optimizes cumulative return rather than immediate spread revenue.

Simulator State vs. Policy Observation

The simulator has a Markov internal state, while the neural policy acts only on an observation of that state.

Simulator state X_t

$X_t = (\text{replay index}, L_t, z_t, x_t, \text{orders}, \text{fills})$

Contains the information needed by the replay environment to generate the next transition.

Markov by construction.

Markov condition: $\Pr(r_t, X_{t+1} \mid \mathcal{H}_t, a_t) = \Pr(r_t, X_{t+1} \mid X_t, a_t).$

Policy observation S_t

$S_t = O(X_t) = \text{Enc}(H_t, \xi_t, u_t)$

H_t : recent LOB history; ξ_t : market signals;
 u_t : agent private state.

Generally $S_t \neq X_t$.

Qualification Bellman value functions are defined on the Markov simulator state X_t ; the neural policy itself faces partial observability through S_t .

Value Functions and the Bellman Relation

The Bellman relation is written for the Markov simulator state X_t ; the policy observes $S_t = O(X_t)$.

$$\begin{aligned} Q_t^\pi(x, a) &= \mathbb{E}_\pi[G_t \mid X_t = x, A_t = a], \\ &= \mathbb{E}[r_t + \gamma V_{t+1}^\pi(X_{t+1}) \mid X_t = x, A_t = a] \end{aligned}$$

State value

$$V_t^\pi(x) = \mathbb{E}_{a \sim \pi_t(\cdot | O(x))} Q_t^\pi(x, a)$$

Advantage

$$A_t^\pi(x, a) = Q_t^\pi(x, a) - V_t^\pi(x)$$

Key point For a fixed policy, V^π averages Q^π . The maximum appears only for the optimal value: $V_t^*(x) = \max_a Q_t^*(x, a)$.

Previous Study: A Discrete Market-Making Agent

The previous agent compressed recent book history and selected one of nine categorical actions.

$$h_t = \left[f_{\psi}(\Phi(L_{t-K+1:t})), z_t, \tau_t \right]$$

Categorical actor

$$\pi_{\theta}(\cdot \mid S_t) = \text{softmax}(g_{\theta}(h_t)).$$

State-value critic

$$V_{\omega}(S_t) = v_{\omega}(h_t).$$

Key point The CNN representation is shared; the actor chooses a category and the critic estimates future return.

Previous Objective: Reward-Based Inventory Control

The previous agent controlled inventory indirectly by penalizing it inside the training reward.

$$r_t^{\text{old}} = r_t^{\text{econ}} - \eta \ell(z_t)$$

$$\pi_\eta \in \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t r_t^{\text{old}} \right]$$

What η controls

A larger η puts stronger pressure on the policy to avoid large inventory.

What η does not specify

The achieved exposure is only observed after training:

$$\eta \longrightarrow \pi_\eta \longrightarrow J_C(\pi_\eta).$$

Key point The penalty controls the strength of the preference against inventory, not the risk level achieved ex ante.

Discrete Quoting: A Fixed Catalogue of Actions

The policy did not choose prices and sizes independently; it selected a predefined template.

$$\mathcal{A}_{\text{old}} = \{\text{cancel_all}\} \cup \{(\kappa_j^b, \kappa_j^a, u_j^b, u_j^a)\}_{j=1}^8$$

Template j becomes executable quotes

$$\tilde{p}_t^b = p_{t,1}^b - \kappa_j^b \Delta, \quad \tilde{p}_t^a = p_{t,1}^a + \kappa_j^a \Delta.$$

Qualification The catalogue contains no bid-only or ask-only action, so direct one-sided inventory correction is unavailable.

PPO: Three Components of the Training Objective

$$\max_{\theta, \omega} \mathcal{L}^{\text{PPO}} = \underbrace{L^{\text{CLIP}}(\theta)}_{\text{actor}} - \underbrace{c_V L_V(\omega)}_{\text{critic}} + \underbrace{c_H \hat{H}(\pi_\theta)}_{\text{exploration}}$$

1. Actor — policy improvement

$$\varrho_t = \frac{\pi_\theta(a_t | S_t)}{\pi_{\theta_{\text{old}}}(a_t | S_t)}$$

$$L^{\text{CLIP}} = \hat{\mathbb{E}}_t \min \left\{ \varrho_t \hat{A}_t, \text{clip}(\varrho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right\}.$$

2. Critic — value regression

$$\hat{V}_t^{\text{target}} = V_{\omega_{\text{old}}}(S_t) + \hat{A}_t, \quad L_V(\omega) = \hat{\mathbb{E}}_t \left[\left(V_\omega(S_t) - \hat{V}_t^{\text{target}} \right)^2 \right].$$

3. Exploration — entropy bonus

$$\hat{H} = -\hat{\mathbb{E}}_t \sum_a \pi_\theta(a | S_t) \log \pi_\theta(a | S_t)$$

Previous Results: Active Quoting and Downside Exposure

The previous PPO agent traded actively, but its PnL distribution retained a substantial negative tail.

Validation: 20 runs $\times$ 20,000 steps

Metric	Reported value
Median PnL	+0.309 USDT
Mean PnL	+0.115 USDT
5% PnL quantile	−1.190 USDT
Quoted rate	0.873

Key point Quoted rate is **87.3%**, yet the lower 5% PnL quantile is **−1.190 USDT**. Activity alone does not control downside exposure.

What the Previous Formulation Leaves Unresolved

Three design choices prevent the old policy from targeting a specified inventory-risk level.

Restriction	Formal consequence
Fixed symmetric catalogue	$\mathcal{A}_{\text{agent}} \subsetneq \mathcal{A}_{\text{baseline}}$
Reward-only risk treatment	η specified, but $J_C(\pi_\eta)$ unknown ex ante
No explicit admissible set	$\Pi_B = \{\pi : J_C(\pi) \leq B\}$ not defined

$$\underbrace{\text{continuous multi-level actions}}_{\text{richer control}} + \underbrace{J_C(\pi) \leq B}_{\text{explicit risk target}}$$

From Discrete Quoting to Risk-Budgeted Market Making

The new study expands the action space and replaces an implicit risk preference with an explicit constraint.

Dimension	Previous study	Current research
Actions	Fixed discrete configurations	Continuous parameters for multi-level quotes
Learning	CNN-PPO	TD3-based imitative learning
Guidance	None	Signal-based expert actions
Risk control	Reward coefficient η	Constraint $J_C(\pi) \leq B$
Estimand	Quoting behavior	Risk response and economic trade-off

Current Problem: Sequential Multi-Level Quoting

The policy produces a structured order set; fills update cash and inventory before the next decision.

$$S_t \xrightarrow{\pi_\theta} a_t \xrightarrow{\text{Decode}} \mathcal{O}_t \xrightarrow{\mathcal{F}} (e_t^b, e_t^a) \xrightarrow{\text{accounting}} (z_{t+1}, x_{t+1})$$

Two outcomes, two objectives

$$(r_t, c_t) = \left(r_t^{\text{econ}}, \left| \frac{z_t}{N_a} \right| \right)$$

Key point Return and inventory exposure are measured separately; the constraint does not hide risk inside the reward.

Historical Replay and Execution Assumptions

Historical replay fixes the external market path; the agent affects only its own hypothetical orders and accounting.

$$P(M_{t+1} \mid \mathcal{H}_t, a_t) = P(M_{t+1} \mid \mathcal{H}_t)$$

Executions are simulated conditionally on the replay

$$(e_t^b, e_t^a) \sim K_{\mathcal{F}}(\cdot \mid \mathcal{O}_t, M_{t:t+1}, U_t), \quad U_{t+1} = g(U_t, \mathcal{O}_t, e_t^b, e_t^a).$$

Qualification All PnL conclusions are conditional on zero market impact and the chosen queue, latency, and fill assumptions.

Market Information and Private State

The policy state combines recent market dynamics with signals and agent-specific information.

$$S_t = \text{Enc}(H_t, \xi_t, u_t), \quad H_t = (L_{t-63}, \dots, L_t)$$

Component	Dimension / horizon	Information
LOB history H_t	depth 5, lookback 64	prices, volumes, book dynamics
Signals ξ_t	$h \in \{10, 60, 120, 300\}$ s	predictive market features
Private state u_t	current time t	inventory and active orders

Key point Every input to S_t must be measurable at decision time t ; future targets may train the encoder but never enter the policy input.

Continuous Parameters for Multi-Level Orders

Four continuous policy outputs are decoded into prices and sizes at several book levels.

$$a_t = (a_{t,1}, \dots, a_{t,4}) \in \mathcal{A} \subseteq \mathbb{R}^4, \quad \mathcal{O}_t = \text{Decode}(S_t, a_t)$$

Structured order collection

$$\begin{aligned} \mathcal{O}_t &= \{(p_{t,\ell}^s, u_{t,\ell}^s) : s \in \{b, a\}, \ell = 1, \dots, M\}, \\ p_{t,\ell}^b &\leq p_{t,1}^b < p_{t,1}^a \leq p_{t,\ell}^a, \quad u_{t,\ell}^s \geq 0. \end{aligned}$$

Key point The decoder guarantees valid quote geometry; its image defines which multi-level orders the policy can express.

Multi-level action representation: Niu et al. (2023), IMM. Action dimension refers to the current research setup.

Inventory Exposure as a Separate Cost

Risk is the expected discounted magnitude of normalized inventory, kept separate from economic return.

$$c_t = \left| \frac{z_t}{N_a} \right|, \quad C_i = (1 - \gamma_C) \sum_{t=0}^{T-1} \gamma_C^t c_{i,t}, \quad J_C(\pi) = \mathbb{E}_\pi[C_i]$$

Current horizon and normalization

$$\gamma_C = 0.999, \quad T = 3600, \quad \sum_{t=0}^{T-1} (1 - \gamma_C) \gamma_C^t = 1 - \gamma_C^T \approx 0.973.$$

Qualification $J_C(\pi) \leq B$ constrains expected exposure, not the inventory of every episode or every time step.

Classical Benchmarks for Inventory-Aware Quoting

Baselines separate inactivity, fixed quoting, heuristic inventory control, and model-based control.

Benchmark	Control role	Formula / implementation
No Trade	no active orders	$u_t^b = u_t^a = 0$
Fixed Quote	constant distances and sizes	$(\delta_t^b, \delta_t^a, u_t^b, u_t^a) = \text{const}$
Inventory-Skew	shift quote center against inventory	center $m_t - g(z_t)$
Avellaneda–Stoikov	reservation price, spread and reference quantities	$p_t^{\text{res}} = m_t - z_t \rho \sigma^2 (T - t),$ $\Delta_t^* = \rho \sigma^2 (T - t) + \frac{2}{\rho} \ln \left(1 + \frac{\rho}{k} \right),$ $p_t^b = p_t^{\text{res}} - \Delta_t^*/2, \quad p_t^a = p_t^{\text{res}} + \Delta_t^*/2,$ $u_t^b = q_t^{b,\text{ref}}, \quad u_t^a = q_t^{a,\text{ref}}.$

Avellaneda and Stoikov (2008), *High-frequency trading in a limit order book*.

Baseline Taxonomy and Experimental Roles

Each comparison answers a different question; only matched branches identify the risk mechanism.

Comparison block	Methods
Market controls	No Trade, Fixed Quote, Inventory-Skew, Avellaneda–Stoikov
Learning controls	LTIIC expert, TD3, IMM
Risk controls	Penalty-IMM (η), RB-IMM (B)

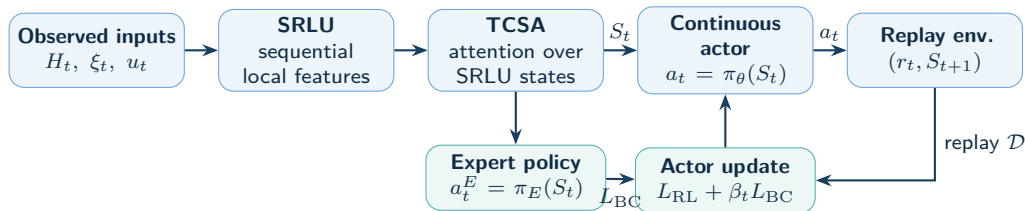
imitation effect : IMM – TD3,

risk mechanism : RB-IMM(B) – Penalty-IMM(η) at matched J_C

Imitative learning reference: Niu et al. (2023), IMM; TD3: Fujimoto et al. (2018).

IMM: Representation, Expert Guidance, and RL

IMM combines predictive representation learning, expert guidance, and continuous-control RL.



What SRLU + TCSA means. SRLU first encodes local limit-order-book patterns; TCSA is applied on top of these SRLU states and selects the most informative time positions and channels for the current decision.

Adapted from Niu et al. (2023), IMM, Figure 3.

Predictive Representation: Information Across Horizons

The encoder is trained to preserve market information useful at several future horizons.

$$\mathcal{H} = \{10, 60, 120, 300\} \text{ s}, \quad \hat{y}_{t,h} = g_h(f_\psi(H_t))$$

Auxiliary training loss

$$L_{\text{pred}}(\psi) = \sum_{h \in \mathcal{H}} \omega_h \mathbb{E}[\ell(\hat{y}_{t,h}, y_{t,h})].$$

Qualification Future data may define training targets $y_{t,h}$, but only time- t information may enter the deployed policy state.

Niu et al. (2023), IMM. Forecast horizons refer to the current research setup.

TD3 Part 1: Why Twin Critics Are Needed

TD3 is the continuous-action backbone: it learns from replay, but explicitly fights Q-value overestimation.

Off-policy critic target

$$\tilde{a}' = \text{clip}_{\mathcal{A}}(\pi_{\theta'}(s') + \epsilon), \quad \epsilon \sim \text{clip}(\mathcal{N}(0, \sigma^2)).$$

$$y = r + \gamma(1 - d) \min_{j=1,2} Q_{\phi'_j}(s', \tilde{a}').$$

Critic regression

$$L_Q(\phi_i) = \mathbb{E}_{\mathcal{D}} (Q_{\phi_i}(s, a) - y)^2, \\ i = 1, 2.$$

Interpretation. The minimum of two target critics is a conservative target: it reduces the tendency to exploit accidentally overestimated actions.

Key point In market making this matters because a tiny optimistic error can push the policy toward aggressive quote sizes or spreads.

TD3 Part 2: Actor Updates and Stabilization

TD3 improves the actor more slowly than the critics, so the policy follows a more reliable value estimate.

Delayed policy update

$$\max_{\theta} \mathbb{E}_{s \sim \mathcal{D}} Q_{\phi_1}(s, \pi_{\theta}(s)),$$
$$\theta \leftarrow \theta \quad \text{only every } d \text{ critic steps.}$$

Soft target networks

$$\phi'_i \leftarrow \tau \phi_i + (1 - \tau) \phi'_i,$$
$$\theta' \leftarrow \tau \theta + (1 - \tau) \theta'.$$

Mechanism	Market-making interpretation
Twin critics	do not trust a quote only because one critic is optimistic
Target smoothing	avoid actions that work only in a tiny neighborhood
Delayed actor	update the policy after critic estimates become less noisy
Soft targets	keep Bellman targets stable across replay updates

Fujimoto, van Hoof and Meger (2018), TD3; Niu et al. (2023), IMM.

Imitation Loss: Expert Guidance During Policy Learning

Expert actions provide a prior for the actor while reinforcement learning remains the optimization target.

$$a^E = \pi_E(S), \quad L_{\text{BC}}(\theta) = \mathbb{E}_{(S, a^E) \sim \mathcal{D}_E} \|\pi_\theta(S) - a^E\|_2^2.$$
$$L_{\text{actor}}^{\text{IMM}} = L_{\text{RL}} + \beta_t L_{\text{BC}}, \quad \beta_{t+1} \leq \beta_t, \quad \beta_t \geq 0.$$

Qualification The expert guides early learning; it does not guarantee optimal actions or final policy quality.

Niu et al. (2023), IMM. Equations show the conceptual imitation objective.

Part II: Explicit Budgets and Completed Evidence

1. **Risk constraint and its Lagrangian**

A precise link to the fixed inventory penalty.

2. **Completed experiments: K0/K4 and D0/D1/D2**

Relative risk, absolute compliance and controller diagnostics.

3. **Future work: distributions and monetary tail risk**

What a quantile critic and a CVaR constraint would add.

The Trading Model

ETHUSDT historical replay; multi-level quoting every 0.5 seconds.

$$X_t \xrightarrow{O} S_t, \quad a_t = \pi_\theta(S_t) \in \mathcal{A} \subseteq \mathbb{R}^4, \quad (X_{t+1}, r_t, c_t) \sim P(\cdot \mid X_t, a_t).$$

X_t	Replay index, LOB, cash, inventory and outstanding orders.
$S_t = O(X_t)$	Encoded market history, signals and private information.
a_t	Four parameters decoded into bid/ask prices and sizes at several levels.
P, ρ_0	Replay transition law and initial-state/window distribution.
θ	Policy parameters; expectation uses the trajectory law induced by ρ_0, P, π_θ .

$T = 3600$ decisions = 30 minutes. The observation S_t need not be Markov.

Model and representation: Part I; IMM, Niu et al. (2023). Research setting: supplied context.

Return, Exposure and the Constrained Objective

Economic objective

$$G_R = \sum_{t=0}^{T-1} \gamma_R^t r_t, \quad J_R(\pi) = \mathbb{E}_\pi[G_R].$$

r_t : economic training reward.

Reported monetary outcome: $P = W_T - W_0$.

Inventory exposure

$$c_t = \frac{|q_t|}{Q_{\text{ref}}}, \quad C_T = (1 - \gamma_C) \sum_{t < T} \gamma_C^t c_t,$$

$$J_C(\pi) = \mathbb{E}_\pi[C_T].$$

Notation bridge: $q = z$, $Q_{\text{ref}} = N_a$ in Part I.

$$\boxed{\max_{\theta} J_R(\pi_{\theta}) \quad \text{subject to} \quad J_C(\pi_{\theta}) \leq B}$$

$B > 0$: predeclared budget; $\gamma_R = \gamma_C = 0.999$ in this study.

Qualification The constraint concerns expected inventory exposure. It does not bound every path, peak inventory or monetary loss.

The Lagrangian: A Price for Expected Exposure

$$\min_{\theta} -J_R(\theta) \quad \text{subject to} \quad J_C(\theta) - B \leq 0.$$

$$\mathcal{L}(\theta, \lambda) = -J_R(\theta) + \lambda(J_C(\theta) - B), \quad \lambda \geq 0$$

Policy minimizes $\mathcal{L}$

Higher return lowers $\mathcal{L}$.

More exposure raises it at price λ .

Multiplier maximizes $\mathcal{L}$

If $J_C > B$, increasing λ penalizes the violation.

If $J_C < B$, its ascent update lowers λ .

Qualification The exact constrained representation is $\inf_{\theta} \sup_{\lambda \geq 0} \mathcal{L}$. A saddle point or convergence is not guaranteed for a neural policy.

How the Multiplier and Policy Interact

$$\mathcal{L}(\theta, \lambda) = -J_R(\theta) + \lambda(J_C(\theta) - B), \quad \lambda \geq 0.$$

Policy: descend in θ

$$\theta_{k+1} = \theta_k - \alpha_k \hat{\nabla}_{\theta} \mathcal{L}(\theta_k, \lambda_k).$$

Seek higher return; weigh exposure by the current λ_k .

Multiplier: ascend in λ

$$\lambda_{k+1} = [\lambda_k + \eta_k(\hat{J}_C - B)]_+.$$

Risk above budget increases its price. Risk below budget decreases it.

$$\hat{J}_C > B \Rightarrow \lambda \uparrow, \quad \hat{J}_C < B \Rightarrow \lambda \downarrow.$$

$[x]_+ = \max(x, 0)$; k indexes optimization iterations; $\alpha_k, \eta_k > 0$ are step sizes.

Idealized static primal–dual update. RCPO: Tessler et al. (2019). Feedback interpretation: Stooke et al. (2020).

The Fixed Penalty Is a Frozen Lagrange Multiplier

For $\gamma_R = \gamma_C = \gamma$:

$$\begin{aligned}\mathcal{L}(\theta, \lambda) &= -\mathbb{E}_\pi \sum_{t < T} \gamma^t r_t + \lambda(1 - \gamma) \mathbb{E}_\pi \sum_{t < T} \gamma^t c_t - \lambda B \\ &= -\mathbb{E}_\pi \sum_{t < T} \gamma^t \underbrace{(r_t - \kappa c_t)}_{r_t^\kappa} - \lambda B, \quad \boxed{\kappa = (1 - \gamma)\lambda}.\end{aligned}$$

- For fixed λ , $-\lambda B$ is constant with respect to the policy.
- Changing B alone cannot change a policy trained with fixed κ .

$$\kappa_{K4} = 0.0131684977, \quad \lambda_0 = \frac{\kappa_{K4}}{1 - 0.999} = 13.1684977.$$

This deck uses $\lambda(J_C - B)$. If the constraint is written as $J_C/B - 1$, its multiplier is $\bar{\lambda} = B\lambda$.

What the Completed Fixed-Penalty Learner Optimizes

$$Q^{\kappa, \pi}(x, a) = \mathbb{E}_{\pi} \left[\sum_{j=0}^{T-t-1} \gamma^j (r_{t+j} - \kappa c_{t+j}) \mid X_t = x, a_t = a \right].$$

$$y^{\kappa} = r - \kappa c + \gamma(1 - d) \min_{i=1,2} \bar{Q}_i^{\kappa}(S', \tilde{a}'), \quad \tilde{a}' = \text{clip}(\bar{\pi}(S') + \epsilon).$$

$$L_{\text{actor}} = -\mathbb{E}_{\mathcal{D}} Q_1^{\kappa}(S, \pi_{\theta}(S)) + \beta_k L_{\text{BC}}, \quad \beta_k \downarrow 0.$$

Arm	Fixed coefficient	Role
K0	$\kappa = 0$	Baseline without the inventory penalty
K4	$\kappa = 0.0131684977$	Frozen strong-penalty policy

d : terminal flag; bars: target networks; $\mathcal{D}$: replay. TD3: Fujimoto et al. (2018); IMM: Niu et al. (2023).

Completed Experiment: A Frozen Paired Protocol

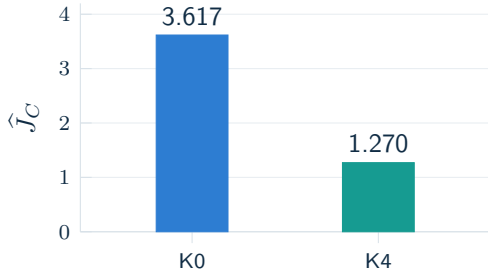
$$\underbrace{5}_{\text{training seeds}} \times \underbrace{2}_{\text{K0 / K4}} \times \underbrace{7}_{\text{matched windows}} = \boxed{70 \text{ evaluation cells}}$$

Checkpoint	Episode 10 for every seed and arm
Risk budgets	$B_{\text{loose}} = 1.214222$; $B_{\text{strict}} = 0.758888$
Activity floor	Filled-notional ratio to K0 at least 0.50
Statistical closure	v0.8a reuses v0.8 outcomes; no new training or rollouts
Selection discipline	No K4 retuning, checkpoint reselection or test access

Seven non-overlapping, time-adjacent 30-minute windows; five seeds: 10801–10805.

Risk Falls by 64.88%; Activity Remains at 75.76%

Mean discounted inventory risk



—64.88%

relative risk reduction

75.76%

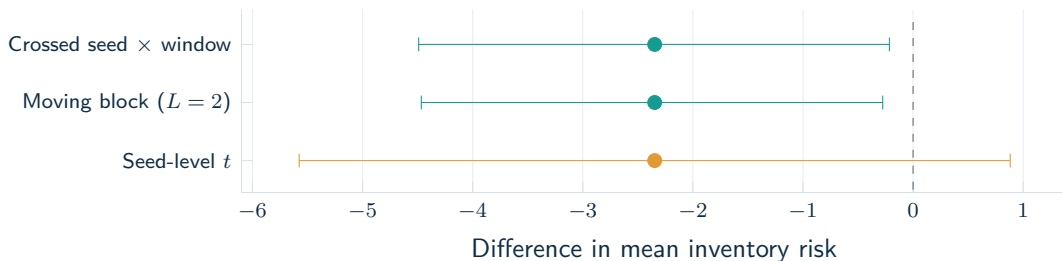
ratio of mean filled notional, K4/K0

42,058.98 / 55,514.81

Activity floor: 50%.

Risk Reduction: Bootstrap Evidence and Seed Uncertainty

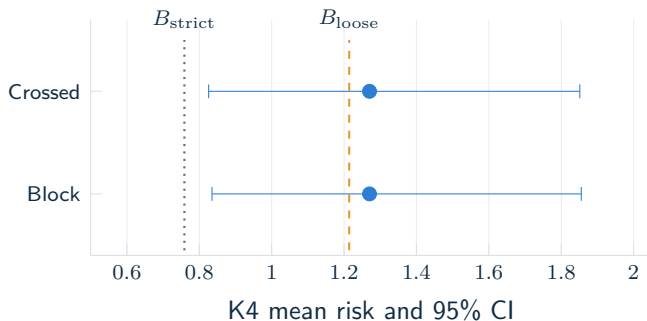
$\Delta_C = J_C(K4) - J_C(K0)$; negative values favor K4. All intervals are 95%.



Crossed: $[-4.491, -0.215]$; block: $[-4.467, -0.275]$; seed t : $[-5.576, 0.882]$.

Qualification Two bootstrap analyses support a relative reduction. Five seeds do not establish full robustness to training randomness.

K4 Does Not Meet the Absolute Budget Criterion



Loose budget

Mean: $1.270 > 1.214$.

Both upper CI bounds exceed B .

Feasible seed means: $3/5$.

Feasible cells: $23/35$.

Key point Relative improvement and absolute compliance are separate empirical claims.

What My Completed Experiment Establishes

Question	Evidence	Conclusion
Relative exposure	−64.88%; two bootstrap CIs below zero	Reduction supported
Activity	Aggregate 75.76%; all five ratios $\geq 50\%$	Floor passed
Absolute budget	Mean 1.270 > 1.214; both upper CIs > B	Criterion failed
Training variability	One reversal; seed t CI crosses zero	Robustness limited
PnL	Wide paired interval includes zero	Unresolved

Key point The fixed-penalty stage is closed. The adaptive experiment tests feedback control of budget expenditure.

Proposed Method: Order-Aware Budget Pacing

1. **A budget path consistent with discounting**

Compare cumulative exposure C_n with an explicit schedule B_n .

2. **Exposure already committed through live orders**

Anticipate inventory that can arise before the next control decision.

3. **Feedback visible to the policy**

Update λ every minute; condition the actor on the budget context.

The specific joint construction is my methodological contribution.

Its incremental empirical value is tested by D2 versus D1.

A Budget Path Consistent with the Discounted Cost

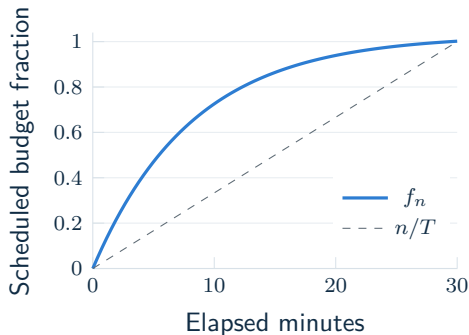
After n completed decisions:

$$f_n = \frac{\sum_{t \leq n} \gamma_C^t}{\sum_{t \leq T} \gamma_C^t} = \frac{1 - \gamma_C^n}{1 - \gamma_C^T},$$

$$\boxed{B_n = B f_n}, \quad B_0 = 0, \quad B_T = B.$$

If expected per-step exposure is constant, its cumulative discounted cost follows the same shape.

$$e_n^{\text{pace}} = \frac{C_n - B_n}{B}, \quad C_n = (1 - \gamma_C) \sum_{t \leq n} \gamma_C^t c_t.$$



Analytical illustration, not a training result. $\gamma_C = 0.999$, $T = 3600$. Code uses $\max(B, 10^{-12})$ in denominators; here $B > 0$.

Exposure in Outstanding Orders

$V_n^{\text{bid}}, V_n^{\text{ask}} \geq 0$: remaining quantities of the agent's active orders.

For any partial fills of those orders, the resulting inventory lies in

$$q'_n \in [q_n - V_n^{\text{ask}}, q_n + V_n^{\text{bid}}].$$

The largest absolute value on this interval is attained at an endpoint:

$$\tilde{c}_n = \frac{\max(|q_n + V_n^{\text{bid}}|, |q_n - V_n^{\text{ask}}|)}{Q_{\text{ref}}}.$$

$$g_n^{\text{order}} = (\tilde{c}_n - c_n)_+, \quad A_n = \frac{g_n^{\text{order}}}{B}.$$

Qualification $\tilde{c}_n$ is an inventory envelope for currently active orders. It is not a fill probability, expected future cost or a trajectory guarantee.

Implemented D2 signal. A_n denotes the order alarm; a_t remains the policy action.

Why Current Inventory Can Miss Committed Exposure

Illustrative order book state; these numbers are not experimental results.

Same current inventory

$q_n = 0, \quad c_n = 0.$	
Live orders	Remaining quantity
Bids	$2Q_{\text{ref}}$
Asks	Q_{ref}

One-sided fills can create inventory.

Potential inventory after fills

$$q'_n \in [-Q_{\text{ref}}, 2Q_{\text{ref}}], \quad \tilde{c}_n = 2.$$

$$g_n^{\text{order}} = 2, \quad A_n = 2/B > 0.$$

If $C_n = B_n$, then $e_n^{\text{pace}} = 0$.

D1 receives zero pacing error.

D2 receives a positive order alarm.

Key point The order signal can raise the risk price before the inventory materializes.

Three Controllers and the Mechanisms They Compare

$$\lambda^+ = \Pi_{[0, \lambda_{\max}]}(\lambda + \eta \text{clip}(u_n, -2, 2)).$$

Arm	Update time	Raw error u_n	Policy context
D0 episodic dual	End of episode	$(C_T - B)/B$	Budget and λ
D1 realized pacing	Every 120 decisions	e_n^{pace}	D0 + remaining budget and pace
D2 order-aware pacing	Every 120 decisions	$e_n^{\text{pace}} + A_n$	D1 + order alarm

$\Pi_{[0, M]}(x) = \min(M, \max(0, x))$. Pacing arms also update at termination.

Controller Scale and the One-Minute Update

$$\lambda_0 = \frac{\kappa_{K4}}{1 - \gamma_C} = 13.1685, \quad \lambda_{\max} = 8\lambda_0 = 105.3480.$$

$$120 \text{ decisions} \times 0.5 \text{ s} = 60 \text{ s}, \quad N_{\text{blocks}} = 30.$$

$$\eta_{D0} = \lambda_0, \quad \eta_{D1,D2} = \frac{\lambda_0}{30} = 0.43895.$$

- λ_0 transfers the scale of K4; it is not another fitted penalty.
- Positive clipped error raises the price; projection keeps it bounded.
- Dividing by 30 sets the update scale, not equivalent controller dynamics.

Qualification Clipping and a finite $\lambda_{\max}$ limit the response. They do not enforce the budget.

The Policy Receives Five Budget Variables

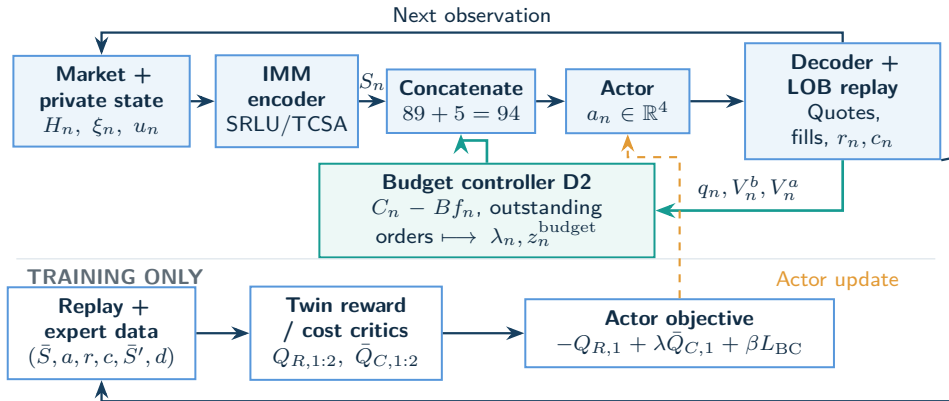
$$z_n^{\text{budget}} = \left(\frac{B}{B_{\text{loose}}}, \quad \frac{B - C_n}{B}, \quad e_n^{\text{pace}}, \quad \frac{\lambda_n}{\lambda_0}, \quad A_n \right).$$

$$\underbrace{S_n}_{89 \text{ features}} = \text{Enc}(H_n, \xi_n, u_n), \quad \underbrace{\bar{S}_n}_{94 \text{ features}} = (S_n, z_n^{\text{budget}}), \quad a_n = \pi_\theta(\bar{S}_n).$$

Remaining fraction	May be negative when the total budget has already been spent.
Feature clipping	Remaining fraction and pace: $[-2, 2]$; alarm: $[0, 2]$; λ/λ_0 : $[0, 8]$.
Arm masks	D0 masks remaining fraction, pace and alarm; D1 masks alarm.

Implemented context: 21 original private features + 5 controller features. The original encoder interface remains unchanged.

RB-IMM Architecture: Policy and Budget Feedback



Teal: budget feedback. Dashed: actor update. Validation uses fixed network weights.

Architecture of the evaluated D2 agent. The controller continues to update its state during evaluation.

Separate Critics for Economic Return and Exposure

Define finite-horizon values on an augmented Markov state $Y_n = (X_n, n, C_n, \lambda_n)$.

$$Q_R^\pi(y, a) = \mathbb{E}_\pi \left[\sum_{j=0}^{T-n-1} \gamma_R^j r_{n+j} \mid Y_n = y, a_n = a \right],$$

$$\bar{Q}_C^\pi(y, a) = (1 - \gamma_C) \mathbb{E}_\pi \left[\sum_{j=0}^{T-n-1} \gamma_C^j c_{n+j} \mid Y_n = y, a_n = a \right].$$

Reward target

$$y_R = r + \gamma_R(1 - d) \min_i Q_{R,i}^-.$$

Normalized cost target

$$y_C = (1 - \gamma_C)c + \gamma_C(1 - d) \min_i \bar{Q}_{C,i}^-.$$

Neural input: $\bar{S}_n$. New normalized cost head: softplus, initialized afresh.

How the Risk Price Changes Actor Learning

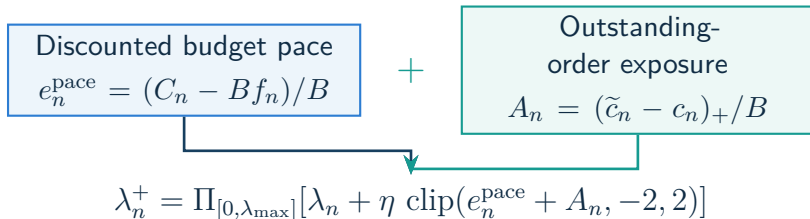
$$L_{\text{actor}}(\theta) = \mathbb{E}_{\mathcal{D}} \left[-Q_{R,1}(\bar{S}, \pi_{\theta}(\bar{S})) + \lambda(\bar{S}) \bar{Q}_{C,1}(\bar{S}, \pi_{\theta}(\bar{S})) \right] + \beta_k L_{\text{BC}}$$

$-Q_R$	Encourages actions with higher predicted economic return.
$\lambda \bar{Q}_C$	Penalizes predicted exposure more strongly when the budget controller raises λ .
$\beta_k L_{\text{BC}}$	Retains early expert guidance; its weight decays.

Qualification For a changing intra-episode λ , this is the implemented local surrogate. The static CMDP Lagrangian motivates it but supplies no convergence or feasibility theorem for D2.

Methodological Contribution

Established foundations: IMM / TD3, constrained RL, feedback multipliers, budget inputs.



One controller, with budget feedback visible to the actor

Empirical test of the order component: $D2 - D1$

Ablation Design: D0, D1 and D2

	D0	D1	D2
Multiplier update	Episode end	Every minute	Every minute
Remaining budget and pace	—	✓	✓
Order alarm	—	—	✓

$D1 - D0$

$D2 - D1$

Pacing and budget context

Added order-aware component

5 seeds

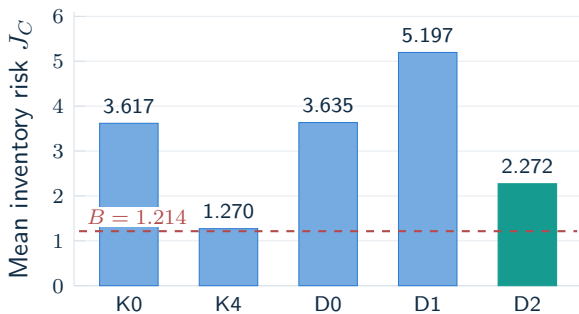
3 arms

7 validation windows

150 training episodes

105 adaptive validation cells

D2 Lowers Mean Risk vs D0/D1, but Misses the Budget



D2 absolute risk

$$1.87 \times B$$

95% intervals:

Crossed: [1.346, 3.574]

Block: [1.360, 3.595]

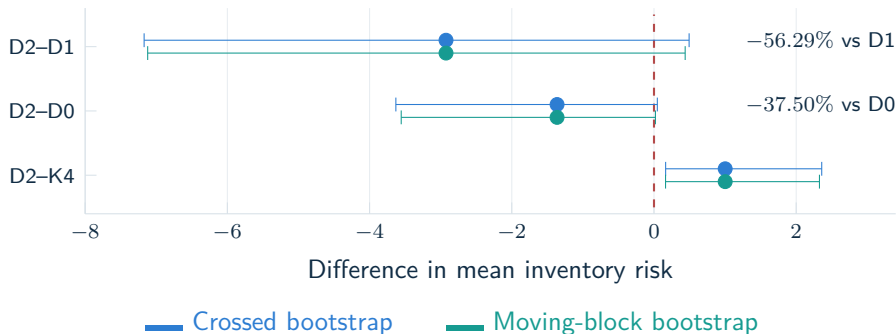
Seed t : [0.871, 3.672]

Only **1 of 5** seed means meets B .

Key point K4 has the lowest mean risk. D2 fails the predeclared absolute-budget criterion.

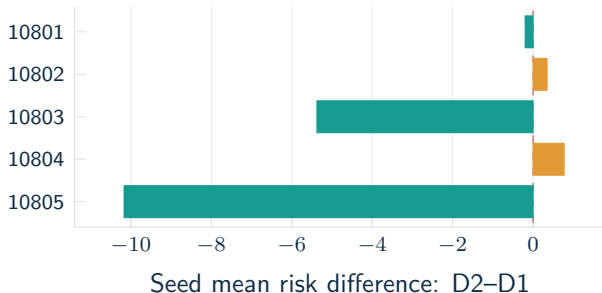
Paired Risk Differences: The D2 Advantage Is Uncertain

Negative differences favor D2. Lines are 95% intervals.



Key point D2-D0 and D2-D1 both include zero. K4 has lower risk under both bootstrap procedures.

The D2-D1 Improvement Concentrates in Two Seeds



Windows favoring D2

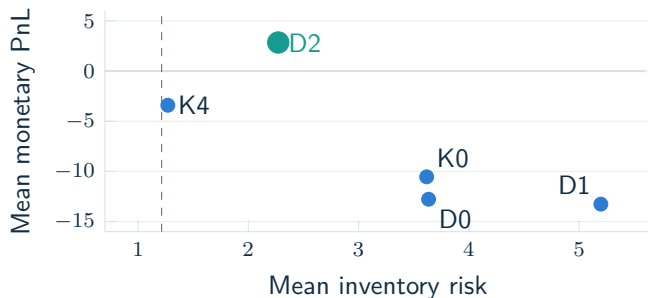
10801	3/7
10802	3/7
10803	7/7
10804	3/7
10805	7/7

Total **23/35**

Seeds 10803 and 10805 drive most of the reduction.

Qualification Matched cells share trained policies and market windows. They are not 35 independent training replications.

D2 Retains Activity and Has Positive Mean PnL



D2 / K0 activity
66.18%

Predeclared floor: 50%

D2 mean PnL
+2.8455

Highest observed mean.

Qualification PnL-difference intervals include zero. Neither superiority nor non-inferiority is established.

All D2 Multipliers End at or Near the Cap

Observed final values

Seed	Final λ
10801	105.347982
10802	105.347982
10803	105.347982
10804	105.347982
10805	105.334483

$$\lambda_{\max} = 105.347982.$$

A mechanism to test

$$u_n = e_n^{\text{pace}} + A_n, \quad A_n \geq 0.$$

Order alarm positive in approximately **88%** of states.

A persistent alarm can push λ upward even near the budget pace.

The cap clips further increases. Negative feedback can still lower λ .

Qualification Endpoint saturation is observed. Its causal role in the budget failure remains a hypothesis.

The Predeclared D2 Acceptance Rule Is Not Met

Criterion	Observed evidence	Status
Absolute budget	Both bootstrap upper endpoints exceed B	FAIL
Trading activity	$66.18\% \geq 50\%$ of K0 filled notional	PASS
Order increment	D2–D1 CIs include zero, and D2 fails its budget gate	FAIL
Order-signal audit	Alarm active in $\sim 88\%$ of states. Paired P95 exposure check not supplied	Partial evidence

Key point The validation experiment is complete. It does not authorize a claim of budget compliance or a robust order-aware advantage.

Next D2 Experiment: Calibrate the Order Feedback

Measure before changing

- Fraction of steps at the cap and time to first saturation.
- Alarm versus subsequent inventory changes and fills.
- Recovery after negative pace errors.

Candidate controlled changes

- Center or calibrate the alarm on training data.
- Weight committed exposure by fill probability.
- Test cap-aware update and recovery rules.

Training-only calibration → freeze → new matched evaluation

Qualification These are hypotheses for a new version. Raising the cap alone has no demonstrated benefit.

Future Work: Equal Mean PnL Can Hide Different Tails

Illustrative distributions only. These numbers are not experimental results.

Policy	Possible PnL outcomes	Mean PnL	Worst 5% mean loss
A	+1 with probability 1	1	-1
B	+2 with probability 0.9 -8 with probability 0.1	1	8

$$\mathbb{E}[P_B] = 0.9(2) + 0.1(-8) = 1 = \mathbb{E}[P_A].$$

For policy B, the worst 5% lies entirely within the 10% probability of losing 8.

Key point A distributional critic represents possible outcomes. CVaR specifies how much their adverse tail matters.

Future work. Distributional RL and CVaR have not been evaluated in v0.9. Distributional RL: Bellemare et al. (2017). CVaR: Rockafellar and Uryasev (2000, 2002).

Distributional RL Learns the Law of Future Return

Scalar value representation

$$Q_P^\pi(y, a) = \mathbb{E}[Z_P^\pi(y, a)].$$

One number summarizes the future monetary outcome.

Proposed distributional critic

$$Z_P^\pi(y, a) \stackrel{d}{=} \sum_{j=0}^{T-n-1} \Delta W_{n+j}.$$

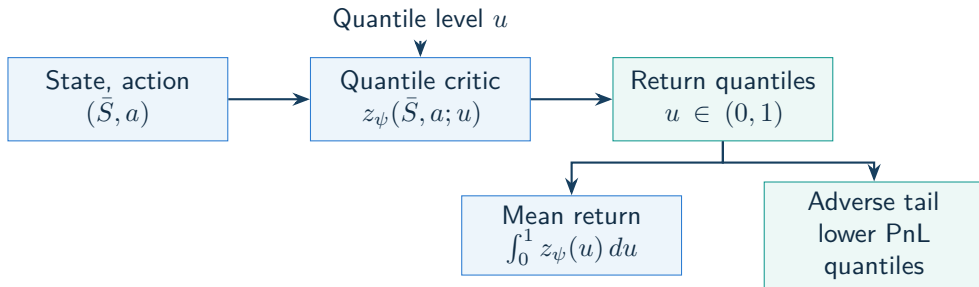
Conditional on $Y_n = y, a_n = a$, then follow π .

$$Z_P^\pi(y, a) \stackrel{d}{=} \Delta W_n + (1 - d_n) Z_P^\pi(Y_{n+1}, \pi(Y_{n+1}))$$

Sample the reward and next state jointly. Draw the continuation from its conditional return law.

Qualification The law describes outcome variability under a fixed policy. Bootstrap CIs describe uncertainty in estimated statistics.

A Quantile Critic Makes the Distribution Usable



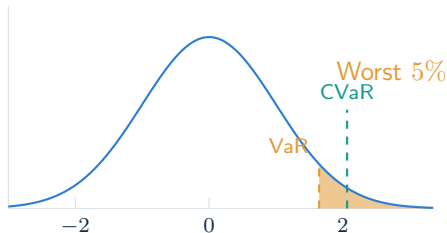
$$t_j = \Delta W + (1 - d) z_{\psi^-}(\bar{S}', \pi^-(\bar{S}'); u_j).$$

Train predicted quantiles against these sampled distributional targets. Keep the inventory-cost critic separate.

Future actor-critic adaptation of quantile RL. Fixed quantiles: Dabney et al. (AAAI 2018). IQN: Dabney et al. (ICML 2018).

CVaR Measures the Severity of the Worst Losses

Use monetary loss $L = -P$. A larger loss is worse.



Standardized loss (illustration)

VaR: the tail threshold

$$q_L(\alpha) = \inf\{\ell : F_L(\ell) \geq \alpha\}.$$

CVaR: the tail average

$$\text{CVaR}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 q_L(u) du.$$

At $\alpha = .95$: mean of the worst 5%. Use fractional mass at atoms.

Key point The confidence level α sets tail size. The monetary budget D sets acceptable tail severity.

Future work. Rockafellar and Uryasev (2000, 2002). Standard normal illustration at $\alpha = .95$.

CVaR Has a Form That Can Be Optimized

$$\text{CVaR}_\alpha(L) = \min_{\eta \in \mathbb{R}} F_\alpha(\eta), \quad F_\alpha(\eta) = \eta + \frac{1}{1-\alpha} \mathbb{E}(L - \eta)_+$$

What the terms mean

η : a candidate loss threshold.

$(L - \eta)_+$: excess loss above it.

$1/(1 - \alpha)$: rescales the tail mass.

Why the optimum is a quantile

For a continuous loss law,

$$F'_\alpha(\eta) = 1 - \frac{\Pr(L > \eta)}{1 - \alpha}.$$

$$F'_\alpha(\eta) = 0 \Rightarrow \Pr(L > \eta) = 1 - \alpha.$$

$$\hat{F}_\alpha(\eta) = \eta + \frac{1}{M(1 - \alpha)} \sum_{m=1}^M (L_m - \eta)_+.$$

Empirical estimate from M episode losses.

Rockafellar–Uryasev representation, 2000 and 2002. Future work. $(x)_+ = \max(x, 0)$.

Future Objective: Two Distinct Risk Constraints

$$\max_{\theta} \mathbb{E}[P_{\theta}] \quad \text{subject to} \quad \underbrace{J_C(\theta) \leq B}_{\text{inventory exposure}}, \quad \underbrace{\text{CVaR}_{\alpha}(-P_{\theta}) \leq D}_{\text{monetary tail loss}}.$$

Introduce η and $F_{\alpha}(\theta, \eta) = \eta + (1 - \alpha)^{-1} \mathbb{E}[(-P_{\theta} - \eta)_+]$.

$$\mathcal{L}_{\text{future}}(\theta, \eta, \lambda, \mu) = -\mathbb{E}[P_{\theta}] + \lambda(J_C(\theta) - B) + \mu(F_{\alpha}(\theta, \eta) - D)$$

Minimize over (θ, η) Maximize over $\lambda, \mu \geq 0$.

Inventory budget B

Expected discounted inventory exposure.

Tail budget D

Severe episode losses, in currency units at a predeclared α .

Qualification Episode CVaR is evaluated under the initial-state mixture. Local conditional CVaR penalties need not optimize this same objective.

Future work. Static episode CVaR. A dynamic risk criterion requires a separate formulation.

Future Tests: Representation and Risk Preference

Step	Change	Question
1	Calibrated D2, mean critics	Can the inventory-budget gate be met reproducibly?
2	Quantile critic, same mean objective	Does modeling the return law improve learning or calibration?
3	Same quantile critic, add CVaR constraint	Does the tail criterion reduce severe losses at an acceptable return cost?

Predeclare α , D , episode horizon and terminal valuation.

Use more episodes and seeds, with matched comparisons and dependence-aware inference.

Key point Distributional learning and CVaR are separate changes. Each needs its own empirical test.

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Clickable article titles link to the primary source.

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